

CSE 542: Big Data Systems, Algorithms and Networks

Assignment 1

By:

(Group 4)

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Requirements of the experiment have been restated here to make sure all the points are completed successfully.



Requirements

When you have completed submission #1, please submit screenshots to show the following tasks have been carried out successfully:

1. A cluster has been created on Cloudlab
2. JDK is setup on each node
3. Passwordless SSH has been setup
4. Yarn has been downloaded in each node
5. Successful setting up of the files core-site.xml, hdfs-site.xml, and yarn-site.xml
6. Setting up of fair-scheduler.xml
7. Creating of slaves file in each node
8. Successfully running HDFS and Yarn (See more in the future slides)
9. Correct number of live nodes on 50070 site

One submission per group is sufficient, however please indicate what each group members is responsible for. Please let me know if you have any questions.

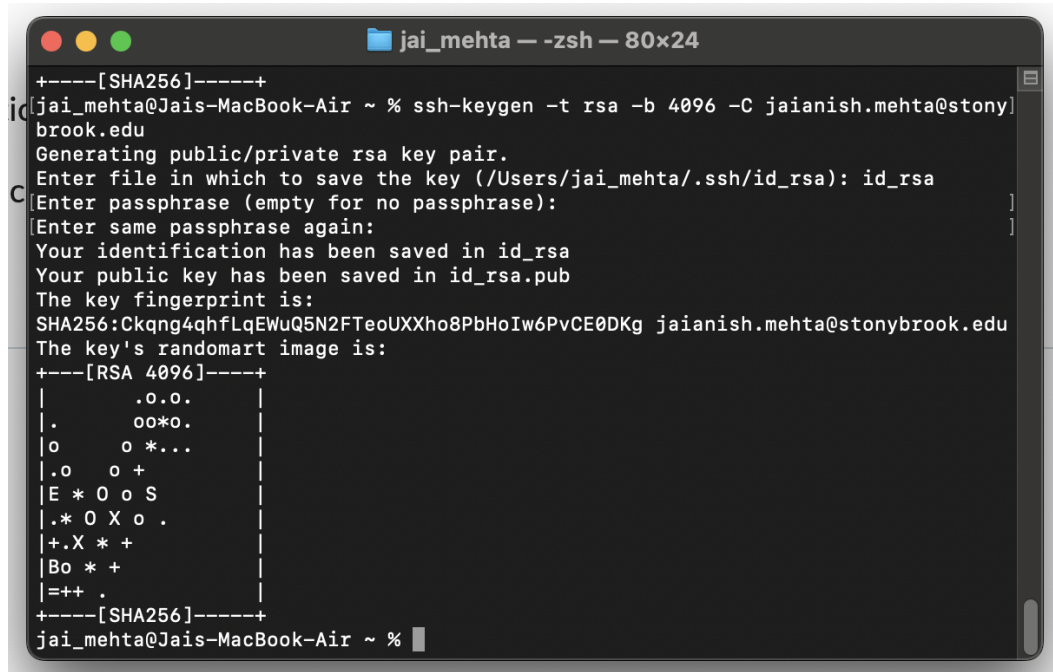
**FAR
BEYOND**

2

Division of work:

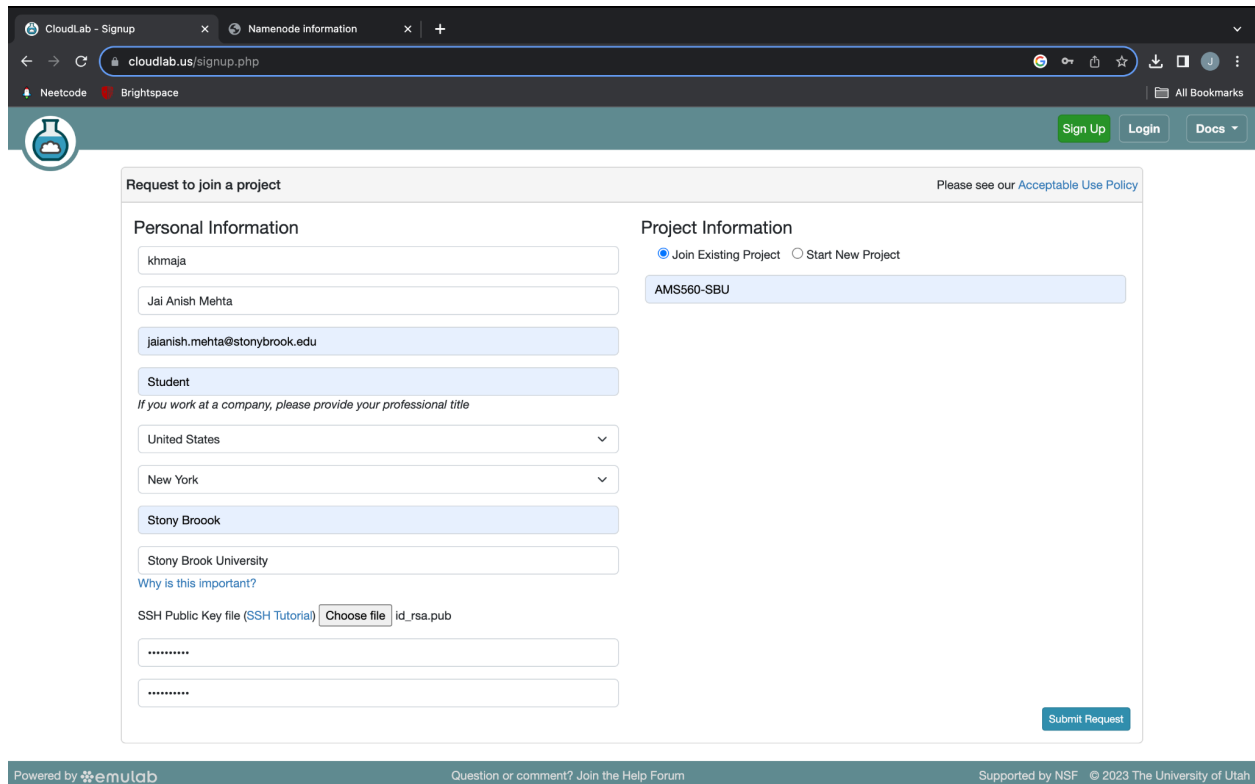
1. Creating the SSH keys and the cluster on Cloud Lab: Jai
2. JDK Setup for ctl: Jai, cp-1: Khush and cp-2: Manas
3. Passwordless SSH setup: Khush
4. Downloading Yarn on ctl: Jai, cp-1: Khush and cp-2: Manas
5. Setting up core-site.xml, hdfs-site.xml and yarn-site.xml on ctl: Jai, cp-1: Khush and cp-2: Manas
6. Setting up fair-scheduler in ctl: Jai, cp-1: Khush and cp-2: Manas
7. Creating slaves file in ctl: Jai, cp-1: Khush and cp-2: Manas
8. Successfully running HDFS, Yarn and jps command: Manas
9. Checking correct number of live nodes: Manas

Creating the SSH key that will be used in the next step. It is stored under the name id_rsa.pub.



```
jai_mehta@Jais-MacBook-Air ~ % ssh-keygen -t rsa -b 4096 -C jaianish.mehta@stonybrook.edu
Generating public/private rsa key pair.
Enter file in which to save the key (/Users/jai_mehta/.ssh/id_rsa): id_rsa
[Enter passphrase (empty for no passphrase): ]
[Enter same passphrase again: ]
Your identification has been saved in id_rsa
Your public key has been saved in id_rsa.pub
The key fingerprint is:
SHA256:Ckqng4qhflqEWuQ5N2FTeoUXXho8PbHoIw6PvCE0DKg jaianish.mehta@stonybrook.edu
The key's randomart image is:
+---[RSA 4096]-----+
|      .O.O.      |
|      OO*O.      |
|     O  O *...   |
|    .O  O +      |
|   E * O o S     |
|  . * O X o .    |
|+.X * +         |
|Bo * +          |
|==++ .          |
+---[SHA256]-----+
jai_mehta@Jais-MacBook-Air ~ %
```

Signing Up on the cloud lab under the existing project AMS560-SBU and adding the SSH public key file.



CloudLab - Signup

cloudlab.us/signup.php

Sign Up Login Docs

Request to join a project [Please see our Acceptable Use Policy](#)

Personal Information

khmaja

Jai Anish Mehta

jaianish.mehta@stonybrook.edu

Student

If you work at a company, please provide your professional title

United States

New York

Stony Brook

Stony Brook University

Why is this important?

SSH Public Key file ([SSH Tutorial](#)) id_rsa.pub

.....

.....

Project Information

☒ Join Existing Project ☐ Start New Project

AMS560-SBU

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Question or comment? Join the Help Forum

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CloudLab - Experiment Status x Namenode information x +

cloudlab.us/status.php?uuid=b235b100-649f-11ee-9f39-e4434b2381fc

Neetcode Brightspace

Experiments Storage News! Docs khmaja

Current Usage: 0 Node Hours, Prev Week: 1.36, Prev Month: 1.36 (30 day rank: 1261 of 1293 users)

▼ Your experiment is ready

Name: hw1group4
 State: ready
 Profile: OpenStack
 RefSpec: refs/heads/master (25f98838)
 Creator: khmaja
 Project: AMS560-SBU
 Created: Oct 6, 2023 7:25 PM
 Started: Oct 6, 2023 7:26 PM
 Expires: Oct 7, 2023 11:25 AM (in 14 hours)

Logs Portal Log Share Save Parameters Modify Create Disk Image Extend Terminate

► Profile Instructions

Topology View List View Manifest Graphs Bindings

ID	Node	Type	Cluster	Status	Startup	Image	SSH command (if you provided your own key)		
ctl	amd155	c6525-25g	Utah	ready	Finished	emulab-ops/UBUNTU22-64-STD	ssh khmaja@amd155.utah.cloudlab.us	☐	
cp-2	amd157	c6525-25g	Utah	ready	Finished	emulab-ops/UBUNTU22-64-STD	ssh khmaja@amd157.utah.cloudlab.us	☐	
cp-1	amd161	c6525-25g	Utah	ready	Finished	emulab-ops/UBUNTU22-64-STD	ssh khmaja@amd161.utah.cloudlab.us	☐	

Not sure how to proceed or have further questions? Join the [users group](#) and ask a question. Be sure to include any error messages in your question and the [URL of your experiment status page](#).

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Starting a new experiment with the profile selected as OpenStack and number of nodes 2.

▼ Please wait while we get your experiment ready

Name: hw1group4
 State: provisioning
 Profile: OpenStack
 RefSpec: refs/heads/master (25f98838)
 Creator: khmaja
 Project: AMS560-SBU
 Started: Oct 6, 2023 5:26 PM
 Expires: Oct 7, 2023 9:25 AM (in 16 hours)

Portal Log Extend Terminate

1. A cluster has been set up consisting of the nodes ctl, cp-1 and cp-2.

 Experiments ▾
 Storage ▾
 News!
 Docs ▾  khmaja ▾

▼ Your experiment is ready (startup services are still running)

Name:	hw1group4
State:	booted (startup services are still running)
Profile:	OpenStack
RefSpec:	refs/heads/master (25f98838)
Creator:	khmaja
Project:	AMS560-SBU
Started:	Oct 6, 2023 5:26 PM
Expires:	Oct 7, 2023 9:25 AM (in 16 hours)

[Share](#) [Save Parameters](#) [Modify](#) [Create Disk Image](#) [Extend](#) [Terminate](#)

▶ Profile Instructions

Topology View | List View | Manifest | Graphs | Bindings

ID ↕	Node ↕	Type ↕	Cluster ↕	Status ↕	Startup ↕	Image ↕	SSH command (if you provided your own key) ↕	☐		
ctl	c220g5-111020	c220g5	Wisc	ready	Running	emulab-ops/UBUNTU22-64-STD	ssh khmaja@c220g5-111020.wisc.cloudlab.us 🔑	☐		
cp-2	c220g5-111001	c220g5	Wisc	ready	Running	emulab-ops/UBUNTU22-64-STD	ssh khmaja@c220g5-111001.wisc.cloudlab.us 🔑	☐		
cp-1	c220g5-111030	c220g5	Wisc	ready	Running	emulab-ops/UBUNTU22-64-STD	ssh khmaja@c220g5-111030.wisc.cloudlab.us 🔑	☐		

Not sure how to proceed or have further questions? Join the [users group](#) and ask a question. Be sure to include any error messages in your question and the [URL of your experiment status page](#).

Install jdk on cp-2

Expires: Oct 7, 2023 11:25 AM (in 16 hours)

```
jai_mehta — khmaja@cp-2: ~ — ssh khmaja@amd157.utah.cloudlab.us — 80...
Processing triggers for libc-bin (2.35-0ubuntu3.4) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for ca-certificates (20230311ubuntu0.22.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
done.
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.42.8+dfsg-1ubuntu0.2) ...
Processing triggers for sgml-base (1.30) ...
Setting up x11proto-dev (2021.5-1) ...
Setting up libxau-dev:amd64 (1:1.0.9-1build5) ...
Setting up libice-dev:amd64 (2:1.0.10-1build2) ...
Setting up libsm-dev:amd64 (2:1.2.3-1build2) ...
Setting up libxdmcp-dev:amd64 (1:1.1.3-0ubuntu5) ...
Setting up libxcb1-dev:amd64 (1.14-3ubuntu3) ...
Setting up libx11-dev:amd64 (2:1.7.5-1ubuntu0.3) ...
Setting up libxt-dev:amd64 (1:1.2.1-1) ...
[khmaja@cp-2:~]$ java -version
openjdk version "1.8.0_382"
OpenJDK Runtime Environment (build 1.8.0_382-8u382-ga-1~22.04.1-b05)
OpenJDK 64-Bit Server VM (build 25.382-b05, mixed mode)
[khmaja@cp-2:~]$
```

```
jai_mehta — khmaja@cp-2: ~ — ssh khmaja@amd157.utah.cloudlab.us — 80...
chmod 0600 ~/.ssh/id_rsa*;
chmod 0600 ~/.ssh/authorized_keys;
rm -rf ~/.ssh/known_hosts;
[echo 'StrictHostKeyChecking no' >> ~/.ssh/config
[khmaja@ctl:~]$ exit
logout
Connection to amd155.utah.cloudlab.us closed.
[jai_mehta@Jais-MacBook-Air ~ % scp ~/.ssh/id_rsa* khmaja@amd157.utah.cloudlab.us]
:~/.ssh/
id_rsa                                100% 3401    58.2KB/s   00:00
id_rsa.pub                           100% 755     12.9KB/s   00:00
[jai_mehta@Jais-MacBook-Air ~ % ssh khmaja@amd157.utah.cloudlab.us]
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Fri Oct 6 23:36:29 2023 from 130.245.192.1
[khmaja@cp-2:~]$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys;
chmod 0600 ~/.ssh/id_rsa*;
chmod 0600 ~/.ssh/authorized_keys;
rm -rf ~/.ssh/known_hosts;
[echo 'StrictHostKeyChecking no' >> ~/.ssh/config
[khmaja@cp-2:~]$
```

Install jdk on cp-1

```
jai_mehta — khmaja@cp-1: ~ — ssh khmaja@amd161.utah.cloudlab.us — 80...
Processing triggers for libc-bin (2.35-0ubuntu3.4) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for ca-certificates (20230311ubuntu0.22.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
done.
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.42.8+dfsg-1ubuntu0.2) ...
Processing triggers for sgml-base (1.30) ...
Setting up x11proto-dev (2021.5-1) ...
Setting up libxau-dev:amd64 (1:1.0.9-1build5) ...
Setting up libice-dev:amd64 (2:1.0.10-1build2) ...
Setting up libsm-dev:amd64 (2:1.2.3-1build2) ...
Setting up libxdmcp-dev:amd64 (1:1.1.3-0ubuntu5) ...
Setting up libxcb1-dev:amd64 (1.14-3ubuntu3) ...
Setting up libx11-dev:amd64 (2:1.7.5-1ubuntu0.3) ...
Setting up libxt-dev:amd64 (1:1.2.1-1) ...
[khmaja@cp-1:~$ java -version
openjdk version "1.8.0_382"
OpenJDK Runtime Environment (build 1.8.0_382-8u382-ga-1~22.04.1-b05)
OpenJDK 64-Bit Server VM (build 25.382-b05, mixed mode)
[khmaja@cp-1:~$
```

3. Setting up passwordless SSH on each node.

```
jai_mehta — khmaja@ctl: ~ — ssh khmaja@amd155.utah.cloudlab.us — 80x...
Setting up libxt-dev:amd64 (1:1.2.1-1) ...
[khmaja@cp-1:~$ java -version
openjdk version "1.8.0_382"
OpenJDK Runtime Environment (build 1.8.0_382-8u382-ga-1~22.04.1-b05)
OpenJDK 64-Bit Server VM (build 25.382-b05, mixed mode)
[khmaja@cp-1:~$ exit
logout
Connection to amd161.utah.cloudlab.us closed.
[jai_mehta@Jais-MacBook-Air ~ % scp ~/.ssh/id_rsa* khmaja@amd155.utah.cloudlab.us:
~/.ssh/
id_rsa                                100% 3401    58.0KB/s   00:00
id_rsa.pub                           100% 755     13.0KB/s   00:00
[jai_mehta@Jais-MacBook-Air ~ % ssh khmaja@amd155.utah.cloudlab.us
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Fri Oct 6 23:30:06 2023 from 130.245.192.1
[khmaja@ctl:~$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys;
chmod 0600 ~/.ssh/id_rsa*;
chmod 0600 ~/.ssh/authorized_keys;
rm -rf ~/.ssh/known_hosts;
echo 'StrictHostKeyChecking no' >> ~/.ssh/config
```

```
jai_mehta — khmaja@cp-1: ~ — ssh khmaja@amd161.utah.cloudlab.us — 80...
chmod 0600 ~/.ssh/id_rsa*;
chmod 0600 ~/.ssh/authorized_keys;
rm -rf ~/.ssh/known_hosts;
echo 'StrictHostKeyChecking no' >> ~/.ssh/config
[khmaja@cp-2:~]$ exit
logout
Connection to amd157.utah.cloudlab.us closed.
jai_mehta@Jais-MacBook-Air ~ % scp ~/.ssh/id_rsa* khmaja@amd161.utah.cloudlab.us:~/.ssh/
id_rsa                                100% 3401    58.0KB/s   00:00
id_rsa.pub                            100% 755     12.9KB/s   00:00
jai_mehta@Jais-MacBook-Air ~ % ssh khmaja@amd161.utah.cloudlab.us
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Fri Oct 6 23:37:56 2023 from 130.245.192.1
[khmaja@cp-1:~]$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys;
chmod 0600 ~/.ssh/id_rsa*;
chmod 0600 ~/.ssh/authorized_keys;
rm -rf ~/.ssh/known_hosts;
echo 'StrictHostKeyChecking no' >> ~/.ssh/config
[khmaja@cp-1:~]$
```

We are able to ssh into each node using the names ctl, cp-1 and cp-2 respectively.

```
jai_mehta — khmaja@ctl: ~ — ssh khmaja@amd155.utah.cloudlab.us — 80x31
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Fri Oct 6 23:50:02 2023 from 130.245.192.1
[khmaja@ctl:~]$ ssh cp-1
Warning: Permanently added 'cp-1' (ECDSA) to the list of known hosts.
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Fri Oct 6 23:46:29 2023 from 130.245.192.1
[khmaja@cp-1:~]$ ssh cp-2
Warning: Permanently added 'cp-2' (ECDSA) to the list of known hosts.
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Fri Oct 6 23:45:21 2023 from 130.245.192.1
[khmaja@cp-2:~]$ ssh ctl
Warning: Permanently added 'ctl' (ECDSA) to the list of known hosts.
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Fri Oct 6 23:51:53 2023 from 130.245.192.1
[khmaja@ctl:~]$
```


4. Downloading yarn on each node and checking it using the ls command.

```
jai_mehta — khmaja@ctl: ~ — ssh khmaja@amd155.utah.cloudlab.us — 80x31
khmaja@ctl:~$ wget https://archive.apache.org/dist/hadoop/core/hadoop-2.7.2/hado
op-2.7.2.tar.gz
--2023-10-06 23:54:45-- https://archive.apache.org/dist/hadoop/core/hadoop-2.7.
2/hadoop-2.7.2.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a
:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... con
nected.
HTTP request sent, awaiting response... 200 OK
Length: 212046774 (202M) [application/x-gzip]
Saving to: 'hadoop-2.7.2.tar.gz'

hadoop-2.7.2.tar.gz 100%[=====>] 202.22M  16.5MB/s   in 13s

2023-10-06 23:54:59 (15.1 MB/s) - 'hadoop-2.7.2.tar.gz' saved [212046774/2120467
74]

khmaja@ctl:~$ tar -xvzf hadoop-2.7.2.tar.gz; mv hadoop-2.7.2 hadoop
hadoop-2.7.2/
hadoop-2.7.2/bin/
hadoop-2.7.2/bin/hdfs
hadoop-2.7.2/bin/mapred.cmd
hadoop-2.7.2/bin/mapred
hadoop-2.7.2/bin/yarn.cmd
hadoop-2.7.2/bin/rcc
hadoop-2.7.2/bin/hdfs.cmd
hadoop-2.7.2/bin/yarn
hadoop-2.7.2/bin/hadoop.cmd
hadoop-2.7.2/bin/hadoop
hadoop-2.7.2/bin/test-container-executor
hadoop-2.7.2/bin/container-executor
```

```
jai_mehta — khmaja@ctl: ~ — ssh khmaja@amd155.utah.cloudlab.us — 80x31
khmaja@ctl:~$ wget https://archive.apache.org/dist/hadoop/core/hadoop-2.7.2/hado
op-2.7.2.tar.gz
--2023-10-06 23:54:45-- https://archive.apache.org/dist/hadoop/core/hadoop-2.7.
2/hadoop-2.7.2.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a
:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... con
nected.
HTTP request sent, awaiting response... 200 OK
Length: 212046774 (202M) [application/x-gzip]
Saving to: 'hadoop-2.7.2.tar.gz'

hadoop-2.7.2.tar.gz 100%[=====>] 202.22M  16.5MB/s   in 13s

2023-10-06 23:54:59 (15.1 MB/s) - 'hadoop-2.7.2.tar.gz' saved [212046774/2120467
74]

khmaja@ctl:~$ tar -xvzf hadoop-2.7.2.tar.gz; mv hadoop-2.7.2 hadoop
hadoop-2.7.2/
hadoop-2.7.2/bin/
hadoop-2.7.2/bin/hdfs
hadoop-2.7.2/bin/mapred.cmd
hadoop-2.7.2/bin/mapred
hadoop-2.7.2/bin/yarn.cmd
hadoop-2.7.2/bin/rcc
hadoop-2.7.2/bin/hdfs.cmd
hadoop-2.7.2/bin/yarn
hadoop-2.7.2/bin/hadoop.cmd
hadoop-2.7.2/bin/hadoop
hadoop-2.7.2/bin/test-container-executor
hadoop-2.7.2/bin/container-executor
```

```
jai_mehta — khmaja@ctl: ~ — ssh khmaja@amd155.utah.cloudlab.us — 80x31
hadoop-2.7.2/share/hadoop/common/lib/api-asn1-api-1.0.0-M20.jar
hadoop-2.7.2/share/hadoop/common/lib/gson-2.2.4.jar
hadoop-2.7.2/share/hadoop/common/lib/commons-cli-1.2.jar
hadoop-2.7.2/share/hadoop/common/lib/junit-4.11.jar
hadoop-2.7.2/share/hadoop/common/lib/jettison-1.1.jar
hadoop-2.7.2/share/hadoop/common/lib/jsr305-3.0.0.jar
hadoop-2.7.2/share/hadoop/common/lib/commons-logging-1.1.3.jar
hadoop-2.7.2/share/hadoop/common/lib/slf4j-log4j12-1.7.10.jar
hadoop-2.7.2/share/hadoop/common/lib/hamcrest-core-1.3.jar
hadoop-2.7.2/share/hadoop/common/lib/slf4j-api-1.7.10.jar
hadoop-2.7.2/share/hadoop/common/lib/commons-httpclient-3.1.jar
hadoop-2.7.2/share/hadoop/common/lib/commons-beanutils-1.7.0.jar
hadoop-2.7.2/share/hadoop/common/lib/paranamer-2.3.jar
hadoop-2.7.2/share/hadoop/common/hadoop-common-2.7.2-tests.jar
hadoop-2.7.2/share/hadoop/common/sources/
hadoop-2.7.2/share/hadoop/common/sources/hadoop-common-2.7.2-sources.jar
hadoop-2.7.2/share/hadoop/common/sources/hadoop-common-2.7.2-test-sources.jar
hadoop-2.7.2/lib/
hadoop-2.7.2/lib/native/
hadoop-2.7.2/lib/native/libhdfs.so
hadoop-2.7.2/lib/native/libhadooputils.a
hadoop-2.7.2/lib/native/libhdfs.so.0.0.0
hadoop-2.7.2/lib/native/libhadoop.so.1.0.0
hadoop-2.7.2/lib/native/libhadoop.a
hadoop-2.7.2/lib/native/libhdfs.a
hadoop-2.7.2/lib/native/libhadoop.so
hadoop-2.7.2/lib/native/libhadooppipes.a
hadoop-2.7.2/LICENSE.txt
khmaja@ctl:~$ ls
hadoop hadoop-2.7.2.tar.gz
khmaja@ctl:~$
```

```
jai_mehta — khmaja@ctl: ~ — ssh khmaja@amd155.utah.cloudlab.us — 80x31
hadoop-2.7.2/share/hadoop/common/sources/hadoop-common-2.7.2-test-sources.jar
hadoop-2.7.2/lib/
hadoop-2.7.2/lib/native/
hadoop-2.7.2/lib/native/libhdfs.so
hadoop-2.7.2/lib/native/libhadooputils.a
hadoop-2.7.2/lib/native/libhdfs.so.0.0.0
hadoop-2.7.2/lib/native/libhadoop.so.1.0.0
hadoop-2.7.2/lib/native/libhadoop.a
hadoop-2.7.2/lib/native/libhdfs.a
hadoop-2.7.2/lib/native/libhadoop.so
hadoop-2.7.2/lib/native/libhadooppipes.a
hadoop-2.7.2/LICENSE.txt
khmaja@cp-2:~$ ls
hadoop hadoop-2.7.2.tar.gz
khmaja@cp-2:~$ ssh ctl
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Fri Oct  6 23:54:02 2023 from 192.168.0.1
khmaja@ctl:~$ export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64/ >> .bashrc
khmaja@ctl:~$ echo export HADOOP_PREFIX=~/.hadoop >> .bashrc;echo export HADOOP_Y
ARN_HOME=~/.hadoop
>> .bashrc;echo export HADOOP_HOME=~/.hadoop >> .bashrc;echo export
HADOOP_CONF_DIR=~/.hadoop/etc/hadoop >> .bashrc;echo export
YARN_CONF_DIR=~/.hadoop/etc/hadoop >> .bashrc;source .bashrc
export HADOOP_YARN_HOME=/users/khmaja/hadoop
export
export
khmaja@ctl:~$
```

```
jai_mehta — khmaja@cp-1: ~ — ssh khmaja@amd155.utah.cloudlab.us — 80...  
hadoop-2.7.2/share/hadoop/common/lib/api-asn1-api-1.0.0-M20.jar  
hadoop-2.7.2/share/hadoop/common/lib/gson-2.2.4.jar  
hadoop-2.7.2/share/hadoop/common/lib/commons-cli-1.2.jar  
hadoop-2.7.2/share/hadoop/common/lib/junit-4.11.jar  
hadoop-2.7.2/share/hadoop/common/lib/jettison-1.1.jar  
hadoop-2.7.2/share/hadoop/common/lib/jsr305-3.0.0.jar  
hadoop-2.7.2/share/hadoop/common/lib/commons-logging-1.1.3.jar  
hadoop-2.7.2/share/hadoop/common/lib/slf4j-log4j12-1.7.10.jar  
hadoop-2.7.2/share/hadoop/common/lib/hamcrest-core-1.3.jar  
hadoop-2.7.2/share/hadoop/common/lib/slf4j-api-1.7.10.jar  
hadoop-2.7.2/share/hadoop/common/lib/commons-httpclient-3.1.jar  
hadoop-2.7.2/share/hadoop/common/lib/commons-beanutils-1.7.0.jar  
hadoop-2.7.2/share/hadoop/common/lib/paranamer-2.3.jar  
hadoop-2.7.2/share/hadoop/common/hadoop-common-2.7.2-tests.jar  
hadoop-2.7.2/share/hadoop/common/sources/  
hadoop-2.7.2/share/hadoop/common/sources/hadoop-common-2.7.2-sources.jar  
hadoop-2.7.2/share/hadoop/common/sources/hadoop-common-2.7.2-test-sources.jar  
hadoop-2.7.2/lib/  
hadoop-2.7.2/lib/native/  
hadoop-2.7.2/lib/native/libhdfs.so  
hadoop-2.7.2/lib/native/libhadooputils.a  
hadoop-2.7.2/lib/native/libhdfs.so.0.0.0  
hadoop-2.7.2/lib/native/libhadoop.so.1.0.0  
hadoop-2.7.2/lib/native/libhadoop.a  
hadoop-2.7.2/lib/native/libhdfs.a  
hadoop-2.7.2/lib/native/libhadoop.so  
hadoop-2.7.2/lib/native/libhadooppipes.a  
hadoop-2.7.2/LICENSE.txt  
khmaja@cp-1:~$ ls  
hadoop hadoop-2.7.2.tar.gz  
khmaja@cp-1:~$
```

```
jai_mehta — khmaja@cp-2: ~ — ssh khmaja@amd155.utah.cloudlab.us — 8...  
hadoop-2.7.2/share/hadoop/common/lib/api-asn1-api-1.0.0-M20.jar  
hadoop-2.7.2/share/hadoop/common/lib/gson-2.2.4.jar  
hadoop-2.7.2/share/hadoop/common/lib/commons-cli-1.2.jar  
hadoop-2.7.2/share/hadoop/common/lib/junit-4.11.jar  
hadoop-2.7.2/share/hadoop/common/lib/jettison-1.1.jar  
hadoop-2.7.2/share/hadoop/common/lib/jsr305-3.0.0.jar  
hadoop-2.7.2/share/hadoop/common/lib/commons-logging-1.1.3.jar  
hadoop-2.7.2/share/hadoop/common/lib/slf4j-log4j12-1.7.10.jar  
hadoop-2.7.2/share/hadoop/common/lib/hamcrest-core-1.3.jar  
hadoop-2.7.2/share/hadoop/common/lib/slf4j-api-1.7.10.jar  
hadoop-2.7.2/share/hadoop/common/lib/commons-httpclient-3.1.jar  
hadoop-2.7.2/share/hadoop/common/lib/commons-beanutils-1.7.0.jar  
hadoop-2.7.2/share/hadoop/common/lib/paranamer-2.3.jar  
hadoop-2.7.2/share/hadoop/common/hadoop-common-2.7.2-tests.jar  
hadoop-2.7.2/share/hadoop/common/sources/  
hadoop-2.7.2/share/hadoop/common/sources/hadoop-common-2.7.2-sources.jar  
hadoop-2.7.2/share/hadoop/common/sources/hadoop-common-2.7.2-test-sources.jar  
hadoop-2.7.2/lib/  
hadoop-2.7.2/lib/native/  
hadoop-2.7.2/lib/native/libhdfs.so  
hadoop-2.7.2/lib/native/libhadooputils.a  
hadoop-2.7.2/lib/native/libhdfs.so.0.0.0  
hadoop-2.7.2/lib/native/libhadoop.so.1.0.0  
hadoop-2.7.2/lib/native/libhadoop.a  
hadoop-2.7.2/lib/native/libhdfs.a  
hadoop-2.7.2/lib/native/libhadoop.so  
hadoop-2.7.2/lib/native/libhadooppipes.a  
hadoop-2.7.2/LICENSE.txt  
[khmaja@cp-2:~$ ls  
hadoop hadoop-2.7.2.tar.gz  
khmaja@cp-2:~$
```



```

hadoop hadoop-2.7.2.tar.gz
[khmaja@cp-2:~$ ssh ctl
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Fri Oct  6 23:54:02 2023 from 192.168.0.1
[khmaja@ctl:~$ export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64/ >> .bashrc
[khmaja@ctl:~$ echo export HADOOP_PREFIX=~/.hadoop >> .bashrc;echo export HADOOP_Y
ARN_HOME=~/.hadoop
>> .bashrc;echo export HADOOP_HOME=~/.hadoop >> .bashrc;echo export
HADOOP_CONF_DIR=~/.hadoop/etc/hadoop >> .bashrc;echo export
YARN_CONF_DIR=~/.hadoop/etc/hadoop >> .bashrc;source .bashrc
export HADOOP_YARN_HOME=/users/khmaja/hadoop
export
export
[khmaja@ctl:~$ cd hadoop/etc/hadoop
[khmaja@ctl:~/hadoop/etc/hadoop$ nano hadoop-env.sh
[khmaja@ctl:~/hadoop/etc/hadoop$

```

```

jai_mehta — khmaja@ctl: ~/hadoop/etc/hadoop — ssh khmaja@amd155.uta...
GNU nano 6.2 hadoop-env.sh *
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd6
# Licensed to the Apache Software Foundation (ASF) under one
# or more contributor license agreements. See the NOTICE file
# distributed with this work for additional information
# regarding copyright ownership. The ASF licenses this file
# to you under the Apache License, Version 2.0 (the
# "License"); you may not use this file except in compliance
# with the License. You may obtain a copy of the License at
#
# http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.
#
# Set Hadoop-specific environment variables here.
#
# The only required environment variable is JAVA_HOME. All others are
# optional. When running a distributed configuration it is best to
# set JAVA_HOME in this file, so that it is correctly defined on
# remote nodes.
#
# The java implementation to use.
export JAVA_HOME=${JAVA_HOME}
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace  ^U Paste     ^J Justify  ^_ Go To Line

```



```
jai_mehta — khmaja@cp-1: ~/hadoop/etc/hadoop — ssh khmaja@amd155.ut...
khmaja@ctl1:~$ export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64/ >> .bashrc
khmaja@ctl1:~$ echo export HADOOP_PREFIX=~/.hadoop >> .bashrc;echo export HADOOP_Y
ARN_HOME=~/.hadoop
>> .bashrc;echo export HADOOP_HOME=~/.hadoop >> .bashrc;echo export
HADOOP_CONF_DIR=~/.hadoop/etc/hadoop >> .bashrc;echo export
YARN_CONF_DIR=~/.hadoop/etc/hadoop >> .bashrc;source .bashrc
export HADOOP_YARN_HOME=/users/khmaja/hadoop
export
export
khmaja@ctl1:~$ cd hadoop/etc/hadoop
khmaja@ctl1:~/hadoop/etc/hadoop$ nano hadoop-env.sh
khmaja@ctl1:~/hadoop/etc/hadoop$ nano hadoop-env.sh
khmaja@ctl1:~/hadoop/etc/hadoop$ ssh cp-1
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Fri Oct  6 23:58:13 2023 from 192.168.0.1
khmaja@cp-1:~$ export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64/ >> .bashrc
echo export HADOOP_PREFIX=~/.hadoop >> .bashrc;echo export HADOOP_YARN_HOME=~/.hadoop
>> .bashrc;echo export HADOOP_HOME=~/.hadoop >> .bashrc;echo export
HADOOP_CONF_DIR=~/.hadoop/etc/hadoop >> .bashrc;echo export
YARN_CONF_DIR=~/.hadoop/etc/hadoop >> .bashrc;source .bashrc
export HADOOP_YARN_HOME=/users/khmaja/hadoop
export
export
khmaja@cp-1:~$ cd hadoop/etc/hadoop
khmaja@cp-1:~/hadoop/etc/hadoop$ nano hadoop-env.sh
```

```
jai_mehta — khmaja@cp-2: ~/hadoop/etc/hadoop — ssh khmaja@amd155.u...
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Fri Oct  6 23:58:13 2023 from 192.168.0.1
khmaja@cp-1:~$ export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64/ >> .bashrc
echo export HADOOP_PREFIX=~/.hadoop >> .bashrc;echo export HADOOP_YARN_HOME=~/.hadoop
>> .bashrc;echo export HADOOP_HOME=~/.hadoop >> .bashrc;echo export
HADOOP_CONF_DIR=~/.hadoop/etc/hadoop >> .bashrc;echo export
YARN_CONF_DIR=~/.hadoop/etc/hadoop >> .bashrc;source .bashrc
export HADOOP_YARN_HOME=/users/khmaja/hadoop
export
export
khmaja@cp-1:~$ cd hadoop/etc/hadoop
khmaja@cp-1:~/hadoop/etc/hadoop$ nano hadoop-env.sh
khmaja@cp-1:~/hadoop/etc/hadoop$ ssh cp-2
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Fri Oct  6 23:59:35 2023 from 192.168.0.7
khmaja@cp-2:~$ export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64/ >> .bashrc
echo export HADOOP_PREFIX=~/.hadoop >> .bashrc;echo export HADOOP_YARN_HOME=~/.hadoop
>> .bashrc;echo export HADOOP_HOME=~/.hadoop >> .bashrc;echo export HADOOP_CONF
_DIR=~/.hadoop/etc/hadoop >> .bashrc;echo export YARN_CONF_DIR=~/.hadoop/etc/hadoop
>> .bashrc;source .bashrc
khmaja@cp-2:~$ cd hadoop/etc/hadoop
khmaja@cp-2:~/hadoop/etc/hadoop$ nano hadoop-env.sh
```

```
jai_mehta — khmaja@ctl: ~/hadoop/etc/hadoop — ssh khmaja@amd155.uta...

* Support:      https://ubuntu.com/advantage
Last login: Sat Oct  7 00:11:03 2023 from 192.168.0.1
khmaja@cp-1:~$ cd hadoop/etc/hadoop
khmaja@cp-1:~/hadoop/etc/hadoop$ nano hadoop-env.sh
khmaja@cp-1:~/hadoop/etc/hadoop$ ssh ctl
Warning: Permanently added 'ctl' (ECDSA) to the list of known hosts.
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:      https://ubuntu.com/advantage
Last login: Sat Oct  7 00:01:41 2023 from 192.168.0.5
khmaja@ctl:~$ cd hadoop/etc/hadoop
khmaja@ctl:~/hadoop/etc/hadoop$ nano hadoop-env.sh
khmaja@ctl:~/hadoop/etc/hadoop$ cd ..
khmaja@ctl:~/hadoop/etc$ cd ..
khmaja@ctl:~/hadoop$ cd ..
khmaja@ctl:~$ sudo mkdir /dev/hdfs; sudo chmod 777 /dev/hdfs
khmaja@ctl:~$ cd hadoop/etc/hadoop/
khmaja@ctl:~/hadoop/etc/hadoop$ ls
capacity-scheduler.xml      httpfs-env.sh              mapred-env.sh
configuration.xml           httpfs-log4j.properties   mapred-queues.xml.template
container-executor.cfg      httpfs-signature.secret   mapred-site.xml.template
core-site.xml               httpfs-site.xml            slaves
hadoop-env.cmd              kms-acls.xml               ssl-client.xml.example
hadoop-env.sh               kms-env.sh                 ssl-server.xml.example
hadoop-metrics2.properties kms-log4j.properties      yarn-env.cmd
hadoop-metrics.properties  kms-site.xml               yarn-env.sh
hadoop-policy.xml           log4j.properties          yarn-site.xml
hdfs-site.xml               mapred-env.cmd
khmaja@ctl:~/hadoop/etc/hadoop$
```

5. Setting up core-site.xml,hdfs-site.xml and yarn-site.xml files.

Set up for node ctl

```
jai_mehta — khmaja@ctl: ~/hadoop/etc/hadoop — ssh khmaja@amd155.uta...
GNU nano 6.2                                core-site.xml *
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  http://www.apache.org/licenses/LICENSE-2.0

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>

<property>
<name>fs.defaultFS</name>
<value>hdfs://ctl:9000</value>
</property>
<property>
<name>io.file.buffer.size</name>
<value>131072</value>
</property>
<property>
<name>hadoop.tmp.dir</name>
<value>/dev/hdfs</value>
</property>

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
jai_mehta — khmaja@ctl: ~/hadoop/etc/hadoop — ssh khmaja@amd155.uta...
GNU nano 6.2                                hdfs-site.xml
Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>

<property>
<name>dfs.replication</name>
<value>1</value>
</property>
<property>
<name>dfs.blocksize</name>
<value>268435456</value>
</property>
<property>
<name>dfs.namenode.handler.count</name>
<value>100</value>
</property>

</configuration>

```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^/ Go To Line

```
jai_mehta — khmaja@ctl: ~/hadoop/etc/hadoop — ssh khmaja@amd155.uta...
GNU nano 6.2                                yarn-site.xml *
airScheduler</value>
</property>
<property>
<name>yarn.scheduler.fair.allocation.file</name>
<value>~/hadoop/etc/fair-scheduler.xml</value>
</property>
<property>
<name>yarn.nodemanager.resource.cpu-vcores</name>
<value>32</value>
</property>
<property>
<name>yarn.nodemanager.resource.memory-mb</name>
<value>65536</value>
</property>
<property>
<name>yarn.nodemanager.hostname</name>
<value>ctl</value>
</property>
<property>
<name>yarn.nodemanager.bind-host</name>
<value>ctl</value>
</property>

</configuration>

```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^/ Go To Line

Set up for node cp-1

```
jai_mehta — khmaja@cp-1: ~/hadoop/etc/hadoop — ssh khmaja@amd155.ut...
GNU nano 6.2                                core-site.xml *

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WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>

<property>
<name>fs.defaultFS</name>
<value>hdfs://ctl:9000</value>
</property>
<property>
<name>io.file.buffer.size</name>
<value>131072</value>
</property>
<property>
<name>hadoop.tmp.dir</name>
<value>/dev/hdfs</value>
</property>

</configuration>

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
jai_mehta — khmaja@cp-1: ~/hadoop/etc/hadoop — ssh khmaja@amd155.ut...
GNU nano 6.2                                hdfs-site.xml

Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>

<property>
<name>dfs.replication</name>
<value>1</value>
</property>
<property>
<name>dfs.blocksize</name>
<value>268435456</value>
</property>
<property>
<name>dfs.namenode.handler.count</name>
<value>100</value>
</property>

</configuration>

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```



```
jai_mehta — khmaja@cp-1: ~/hadoop/etc/hadoop — ssh khmaja@amd155.ut...
GNU nano 6.2                                yarn-site.xml *
<property>
<name>yarn.resourcemanager.scheduler.class</name>
<value>org.apache.hadoop.yarn.server.resourcemanager.scheduler.fair.F
airScheduler</value>
</property>
<property>
<name>yarn.scheduler.fair.allocation.file</name>
<value>~/hadoop/etc/fair-scheduler.xml</value>
</property>
<property>
<name>yarn.nodemanager.resource.cpu-vcores</name>
<value>32</value>
</property>
<property>
<name>yarn.nodemanager.resource.memory-mb</name>
<value>65536</value>
</property>
<property>
<name>yarn.nodemanager.hostname</name>
<value> cp-1</value>
</property>
<property>
<name>yarn.nodemanager.bind-host</name>
<value> cp-1</value>
</property>
</configuration>

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^/ Go To Line
```

Set up for node cp-2

```
jai_mehta — khmaja@cp-2: ~/hadoop/etc/hadoop — ssh khmaja@amd155.u...
GNU nano 6.2                                core-site.xml *
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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>

<property>
<name>fs.defaultFS</name>
<value>hdfs://ctl:9000</value>
</property>
<property>
<name>io.file.buffer.size</name>
<value>131072</value>
</property>
<property>
<name>hadoop.tmp.dir</name>
<value>/dev/hdfs</value>
</property>

</configuration>


```

```
jai_mehta — khmaja@cp-2: ~/hadoop/etc/hadoop — ssh khmaja@amd155.u...
GNU nano 6.2                                hdfs-site.xml *
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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
<name>dfs.replication</name>
<value>1</value>
</property>
<property>
<name>dfs.blocksize</name>
<value>268435456</value>
</property>
<property>
<name>dfs.namenode.handler.count</name>
<value>100</value>
</property>
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
jai_mehta — khmaja@cp-2: ~/hadoop/etc/hadoop — ssh khmaja@amd155.u...
GNU nano 6.2                                yarn-site.xml *
<property>
<name>yarn.resourcemanager.scheduler.class</name>
<value>org.apache.hadoop.yarn.server.resourcemanager.scheduler.fair.F
airScheduler</value>
</property>
<property>
<name>yarn.scheduler.fair.allocation.file</name>
<value>~/hadoop/etc/fair-scheduler.xml</value>
</property>
<property>
<name>yarn.nodemanager.resource.cpu-vcores</name>
<value>32</value>
</property>
<property>
<name>yarn.nodemanager.resource.memory-mb</name>
<value>65536</value>
</property>
<property>
<name>yarn.nodemanager.hostname</name>
<value> cp-2</value>
</property>
<property>
<name>yarn.nodemanager.bind-host</name>
<value> cp-2</value>
</property>
</configuration>
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

6. Setting up of fair-scheduler.xml

Set up for ctl

```
jai_mehta — khmaja@ctl: ~/hadoop/etc — ssh khmaja@amd155.utah.cloudl...

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/advantage
Last login: Sat Oct  7 00:01:41 2023 from 192.168.0.5
[khmaja@ctl:~]$ cd hadoop/etc/hadoop
[khmaja@ctl:~/hadoop/etc/hadoop$ nano hadoop-env.sh
[khmaja@ctl:~/hadoop/etc/hadoop$ cd ..
[khmaja@ctl:~/hadoop/etc$ cd ..
[khmaja@ctl:~/hadoop$ cd ..
[khmaja@ctl:~]$ sudo mkdir /dev/hdfs; sudo chmod 777 /dev/hdfs
[khmaja@ctl:~]$ cd hadoop/etc/hadoop/
[khmaja@ctl:~/hadoop/etc/hadoop$ ls
capacity-scheduler.xml      httpfs-env.sh              mapred-env.sh
configuration.xml           httpfs-log4j.properties   mapred-queues.xml.template
container-executor.cfg      httpfs-signature.secret   mapred-site.xml.template
core-site.xml               httpfs-site.xml            slaves
hadoop-env.cmd              kms-acls.xml               ssl-client.xml.example
hadoop-env.sh               kms-env.sh                 ssl-server.xml.example
hadoop-metrics2.properties kms-log4j.properties      yarn-env.cmd
hadoop-metrics.properties  kms-site.xml               yarn-env.sh
hadoop-policy.xml           log4j.properties          yarn-site.xml
hdfs-site.xml               mapred-env.cmd
[khmaja@ctl:~/hadoop/etc/hadoop$ nano core-site.xml
[khmaja@ctl:~/hadoop/etc/hadoop$ nano hdfs-site.xml
[khmaja@ctl:~/hadoop/etc/hadoop$ nano hdfs-site.xml
[khmaja@ctl:~/hadoop/etc/hadoop$ nano yarn-site.xml
[khmaja@ctl:~/hadoop/etc/hadoop$ nano slaves
[khmaja@ctl:~/hadoop/etc/hadoop$ cd ..
[khmaja@ctl:~/hadoop/etc$ touch fair-scheduler.xml
[khmaja@ctl:~/hadoop/etc$
```

```
jai_mehta — khmaja@ctl: ~/hadoop/etc — ssh khmaja@amd155.utah.cloudl...
GNU nano 6.2                                fair-scheduler.xml *
<?xml version="1.0"?>
<allocations>
<defaultQueueSchedulingPolicy>drf</
defaultQueueSchedulingPolicy>
<queue name="queue0">
<weight>1</weight>
<allowPreemptionFrom>false</allowPreemptionFrom>
<schedulingPolicy>fifo</schedulingPolicy>
</queue>
<queue name="queue1">
<weight>1</weight>
<allowPreemptionFrom>true</allowPreemptionFrom>
</queue>
</allocations>

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Set up for cp-1

```
jai_mehta — khmaja@cp-1: ~/hadoop/etc — ssh khmaja@amd155.utah.clou...

khmaja@cp-1:~/hadoop/etc/hadoop$ cd ..
khmaja@cp-1:~/hadoop/etc$ cd ..
khmaja@cp-1:~/hadoop$ cd ..
khmaja@cp-1:~$ ls
hadoop  hadoop-2.7.2.tar.gz
khmaja@cp-1:~$ sudo mkdir /dev/hdfs; sudo chmod 777 /dev/hdfs
khmaja@cp-1:~$ cd hadoop/etc/hadoop/
khmaja@cp-1:~/hadoop/etc/hadoop$ nano co
configuration.xml      container-executor.cfg  core-site.xml
khmaja@cp-1:~/hadoop/etc/hadoop$ ls
capacity-scheduler.xml  httpfs-env.sh           mapred-env.sh
configuration.xml       httpfs-log4j.properties mapred-queues.xml.template
container-executor.cfg  httpfs-signature.secret mapred-site.xml.template
core-site.xml           httpfs-site.xml         slaves
hadoop-env.cmd          kms-acls.xml            ssl-client.xml.example
hadoop-env.sh           kms-env.sh              ssl-server.xml.example
hadoop-metrics2.properties kms-log4j.properties  yarn-env.cmd
hadoop-metrics.properties kms-site.xml            yarn-env.sh
hadoop-policy.xml       log4j.properties       yarn-site.xml
hdfs-site.xml           mapred-env.cmd
khmaja@cp-1:~/hadoop/etc/hadoop$ nano core-site.xml
khmaja@cp-1:~/hadoop/etc/hadoop$ nano hdfs-site.xml
khmaja@cp-1:~/hadoop/etc/hadoop$ nano hdfs-site.xml
khmaja@cp-1:~/hadoop/etc/hadoop$ nano yarn-site.xml
khmaja@cp-1:~/hadoop/etc/hadoop$ nano yarn-site.xml
khmaja@cp-1:~/hadoop/etc/hadoop$ nano slaves
khmaja@cp-1:~/hadoop/etc/hadoop$ cd ..
khmaja@cp-1:~/hadoop/etc$ touch fair-scheduler.xml
khmaja@cp-1:~/hadoop/etc$ nano fair-scheduler.xml
khmaja@cp-1:~/hadoop/etc$
```

```
jai_mehta — khmaja@cp-1: ~/hadoop/etc — ssh khmaja@amd155.utah.clou...

GNU nano 6.2                fair-scheduler.xml *
<?xml version="1.0"?>
<allocations>
<defaultQueueSchedulingPolicy>drf</
defaultQueueSchedulingPolicy>
<queue name="queue0">
<weight>1</weight>
<allowPreemptionFrom>false</allowPreemptionFrom>
<schedulingPolicy>fifo</schedulingPolicy>
</queue>
<queue name="queue1">
<weight>1</weight>
<allowPreemptionFrom>true</allowPreemptionFrom>
</queue>
</allocations>

```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

Set up for cp-2

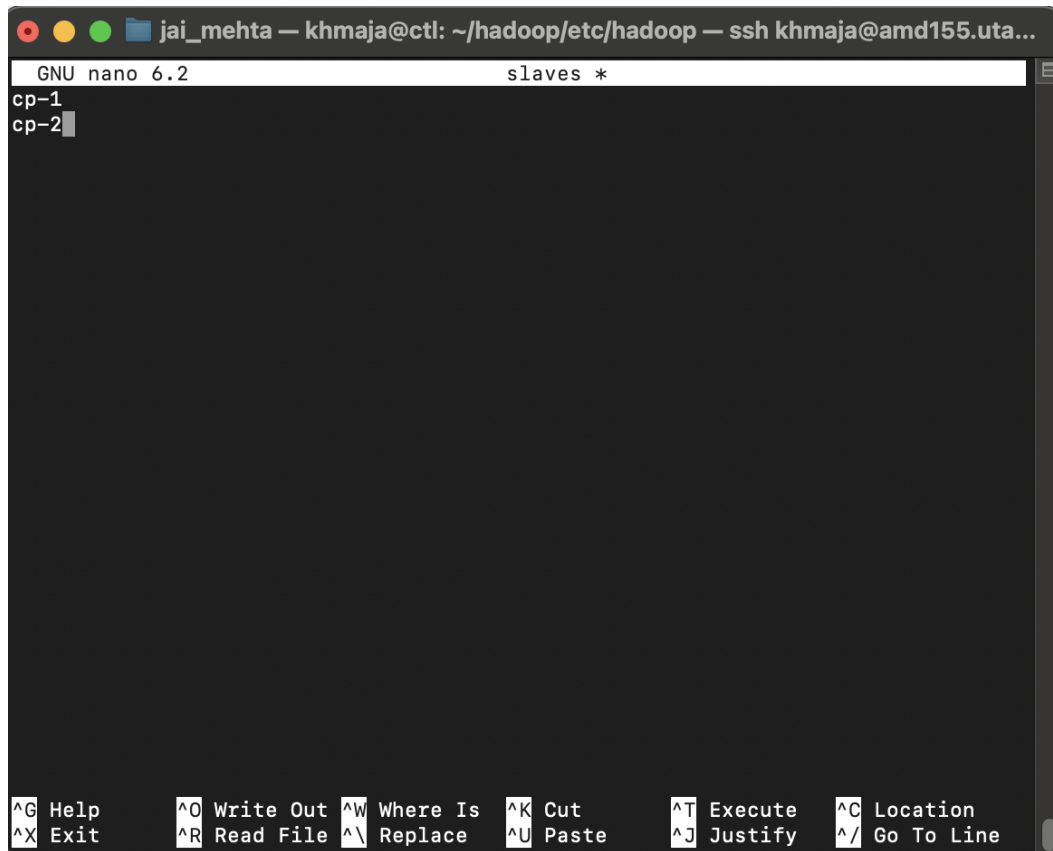
```
jai_mehta — khmaja@cp-2: ~/hadoop/etc — ssh khmaja@amd155.utah.clou...
khmaja@cp-1:~/hadoop/etc/hadoop$ ssh cp-2
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Sat Oct  7 00:14:43 2023 from 192.168.0.7
khmaja@cp-2:~$ cd hadoop/etc/hadoop/
khmaja@cp-2:~/hadoop/etc/hadoop$ sudo mkdir /dev/hdfs; sudo chmod 777 /dev/hdfs
khmaja@cp-2:~/hadoop/etc/hadoop$ ls
capacity-scheduler.xml      httpfs-env.sh              mapred-env.sh
configuration.xsl          httpfs-log4j.properties   mapred-queues.xml.template
container-executor.cfg     httpfs-signature.secret   mapred-site.xml.template
core-site.xml              httpfs-site.xml           slaves
hadoop-env.cmd             kms-acls.xml              ssl-client.xml.example
hadoop-env.sh              kms-env.sh                ssl-server.xml.example
hadoop-metrics2.properties kms-log4j.properties     yarn-env.cmd
hadoop-metrics.properties kms-site.xml              yarn-env.sh
hadoop-policy.xml          log4j.properties         yarn-site.xml
hdfs-site.xml              mapred-env.cmd
khmaja@cp-2:~/hadoop/etc/hadoop$ nano core-site.xml
khmaja@cp-2:~/hadoop/etc/hadoop$ nano hdfs-site.xml
khmaja@cp-2:~/hadoop/etc/hadoop$ nano yarn-site.xml
khmaja@cp-2:~/hadoop/etc/hadoop$ nano slaves
khmaja@cp-2:~/hadoop/etc/hadoop$ cd ..
khmaja@cp-2:~/hadoop/etc$ touch fair-scheduler.xml
khmaja@cp-2:~/hadoop/etc$ nano fair-scheduler.xml
```

```
jai_mehta — khmaja@cp-2: ~/hadoop/etc — ssh khmaja@amd155.utah.clou...
GNU nano 6.2                                fair-scheduler.xml *
<?xml version="1.0"?>
<allocations>
<defaultQueueSchedulingPolicy>drf</
defaultQueueSchedulingPolicy>
<queue name="queue0">
<weight>1</weight>
<allowPreemptionFrom>false</allowPreemptionFrom>
<schedulingPolicy>fifo</schedulingPolicy>
</queue>
<queue name="queue1">
<weight>1</weight>
<allowPreemptionFrom>true</allowPreemptionFrom>
</queue>
</allocations>
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

7. Creating slaves file in each node

Creating in ctl

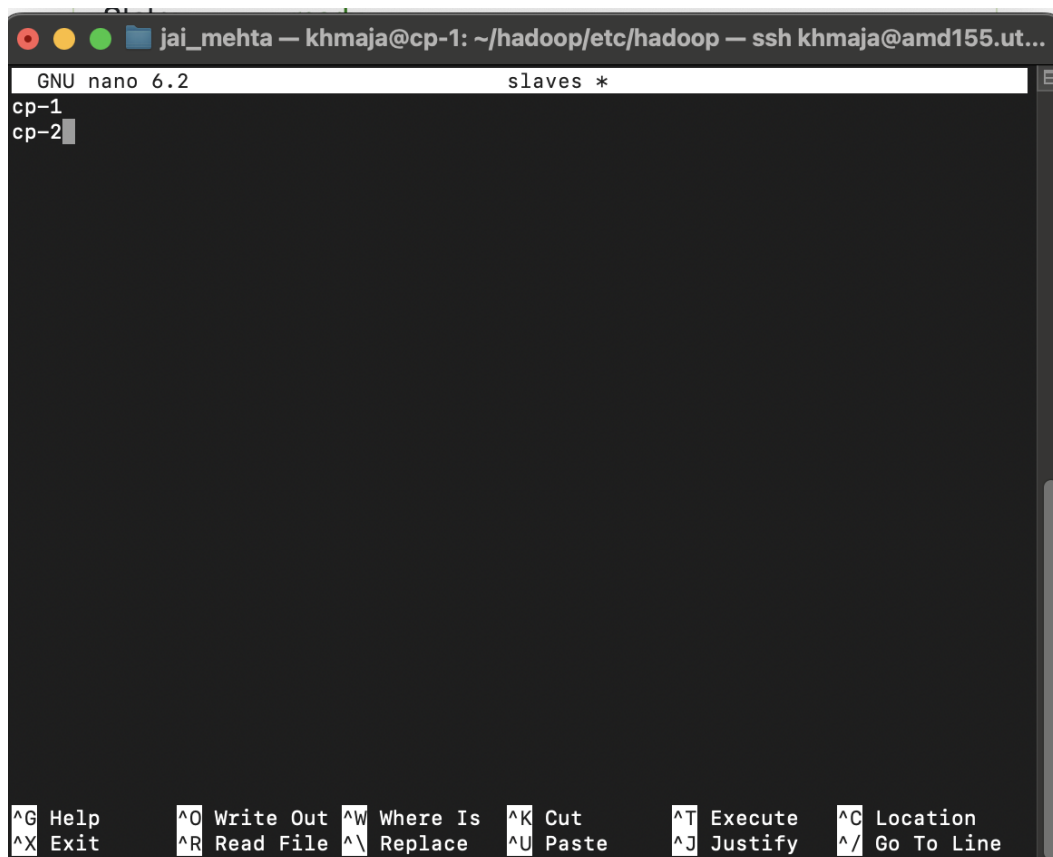


A terminal window titled "jai_mehta — khmaja@ctl: ~/hadoop/etc/hadoop — ssh khmaja@amd155.uta...". The window shows the GNU nano 6.2 editor editing a file named "slaves *". The cursor is on the second line, which contains "cp-2". The first line contains "cp-1". The bottom of the window displays a series of keyboard shortcuts for nano, such as ^G Help, ^O Write Out, ^W Where Is, ^K Cut, ^T Execute, ^C Location, ^X Exit, ^R Read File, ^\ Replace, ^U Paste, ^J Justify, and ^/ Go To Line.

```
GNU nano 6.2          slaves *
cp-1
cp-2
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

Creating slaves file in cp-1



A terminal window titled "jai_mehta — khmaja@cp-1: ~/hadoop/etc/hadoop — ssh khmaja@amd155.uta...". The window shows the GNU nano 6.2 editor editing a file named "slaves *". The cursor is on the second line, which contains "cp-2". The first line contains "cp-1". The bottom of the window displays a series of keyboard shortcuts for nano, such as ^G Help, ^O Write Out, ^W Where Is, ^K Cut, ^T Execute, ^C Location, ^X Exit, ^R Read File, ^\ Replace, ^U Paste, ^J Justify, and ^/ Go To Line.

```
GNU nano 6.2          slaves *
cp-1
cp-2
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

jai_mehta — khmaja@cp-1: ~/hadoop/etc — ssh khmaja@amd155.utah.clou...

```
khmaja@cp-1:~/hadoop/etc/hadoop$ cd ..
khmaja@cp-1:~/hadoop/etc$ cd ..
khmaja@cp-1:~/hadoop$ cd ..
khmaja@cp-1:~$ ls
hadoop  hadoop-2.7.2.tar.gz
khmaja@cp-1:~$ sudo mkdir /dev/hdfs; sudo chmod 777 /dev/hdfs
khmaja@cp-1:~$ cd hadoop/etc/hadoop/
khmaja@cp-1:~/hadoop/etc/hadoop$ nano co
configuration.xml      container-executor.cfg  core-site.xml
khmaja@cp-1:~/hadoop/etc/hadoop$ ls
capacity-scheduler.xml  httpfs-env.sh          mapred-env.sh
configuration.xml       httpfs-log4j.properties mapred-queues.xml.template
container-executor.cfg  httpfs-signature.secret mapred-site.xml.template
core-site.xml           httpfs-site.xml        slaves
hadoop-env.cmd          kms-acls.xml            ssl-client.xml.example
hadoop-env.sh           kms-env.sh              ssl-server.xml.example
hadoop-metrics2.properties kms-log4j.properties  yarn-env.cmd
hadoop-metrics.properties kms-site.xml            yarn-env.sh
hadoop-policy.xml       log4j.properties       yarn-site.xml
hdfs-site.xml           mapred-env.cmd
khmaja@cp-1:~/hadoop/etc/hadoop$ nano core-site.xml
khmaja@cp-1:~/hadoop/etc/hadoop$ nano hdfs-site.xml
khmaja@cp-1:~/hadoop/etc/hadoop$ nano hdfs-site.xml
khmaja@cp-1:~/hadoop/etc/hadoop$ nano yarn-site.xml
khmaja@cp-1:~/hadoop/etc/hadoop$ nano yarn-site.xml
khmaja@cp-1:~/hadoop/etc/hadoop$ nano slaves
khmaja@cp-1:~/hadoop/etc/hadoop$ cd ..
khmaja@cp-1:~/hadoop/etc$
```

Creating Slaves file in cp-2

```
jai_mehta — khmaja@cp-2: ~/hadoop/etc/hadoop — ssh khmaja@amd155.u...
GNU nano 6.2                               slaves *
cp-1
cp-2
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

```
jai_mehta — khmaja@cp-2: ~/hadoop/etc/hadoop — ssh khmaja@amd155.u...
khmaja@cp-1:~/hadoop/etc/hadoop$ ssh cp-2
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-69-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Sat Oct  7 00:14:43 2023 from 192.168.0.7
khmaja@cp-2:~$ cd hadoop/etc/hadoop/
khmaja@cp-2:~/hadoop/etc/hadoop$ sudo mkdir /dev/hdfs; sudo chmod 777 /dev/hdfs
khmaja@cp-2:~/hadoop/etc/hadoop$ ls
capacity-scheduler.xml  httpfs-env.sh          mapred-env.sh
configuration.xml       httpfs-log4j.properties mapred-queues.xml.template
container-executor.cfg  httpfs-signature.secret mapred-site.xml.template
core-site.xml           httpfs-site.xml        slaves
hadoop-env.cmd          kms-acls.xml           ssl-client.xml.example
hadoop-env.sh           kms-env.sh             ssl-server.xml.example
hadoop-metrics2.properties kms-log4j.properties  yarn-env.cmd
hadoop-metrics.properties kms-site.xml           yarn-env.sh
hadoop-policy.xml       log4j.properties      yarn-site.xml
hdfs-site.xml           mapred-env.cmd
khmaja@cp-2:~/hadoop/etc/hadoop$ nano core-site.xml
khmaja@cp-2:~/hadoop/etc/hadoop$ nano hdfs-site.xml
khmaja@cp-2:~/hadoop/etc/hadoop$ nano yarn-site.xml
khmaja@cp-2:~/hadoop/etc/hadoop$ nano slaves
khmaja@cp-2:~/hadoop/etc/hadoop$
```


8. Successfully running HDFS and Yarn

```
jai_mehta — khmaja@ctl: ~ — ssh khmaja@amd155.utah.cloudlab.us — 80x31
23/10/07 01:26:06 INFO namenode.NNStorageRetentionManager: Going to retain 1 ima
ges with txid >= 0
23/10/07 01:26:06 INFO util.ExitUtil: Exiting with status 0
23/10/07 01:26:06 INFO namenode.NameNode: SHUTDOWN_MSG:
/*****
SHUTDOWN_MSG: Shutting down NameNode at ctl.hw1group4.ams560-sbu-pg0.utah.cloudl
ab.us/128.110.219.66
*****/
khmaja@ctl:~$ hadoop/sbin/start-dfs.sh
Starting namenodes on [ctl]
ctl: starting namenode, logging to /users/khmaja/hadoop/logs/hadoop-khmaja-namen
ode-ctl.hw1group4.ams560-sbu-pg0.utah.cloudlab.us.out
cp-2: Warning: Permanently added 'cp-2' (ECDSA) to the list of known hosts.
cp-2: starting datanode, logging to /users/khmaja/hadoop/logs/hadoop-khmaja-data
node-cp-2.hw1group4.ams560-sbu-pg0.utah.cloudlab.us.out
cp-1: starting datanode, logging to /users/khmaja/hadoop/logs/hadoop-khmaja-data
node-cp-1.hw1group4.ams560-sbu-pg0.utah.cloudlab.us.out
Starting secondary namenodes [0.0.0.0]
0.0.0.0: Warning: Permanently added '0.0.0.0' (ECDSA) to the list of known hosts
.
0.0.0.0: starting secondarynamenode, logging to /users/khmaja/hadoop/logs/hadoop
-khmaja-secondarynamenode-ctl.hw1group4.ams560-sbu-pg0.utah.cloudlab.us.out
khmaja@ctl:~$ hadoop/sbin/start-yarn.sh
starting yarn daemons
starting resourcemanager, logging to /users/khmaja/hadoop/logs/yarn-khmaja-resou
rcemanager-ctl.hw1group4.ams560-sbu-pg0.utah.cloudlab.us.out
cp-1: starting nodemanager, logging to /users/khmaja/hadoop/logs/yarn-khmaja-nod
emanager-cp-1.hw1group4.ams560-sbu-pg0.utah.cloudlab.us.out
cp-2: starting nodemanager, logging to /users/khmaja/hadoop/logs/yarn-khmaja-nod
emanager-cp-2.hw1group4.ams560-sbu-pg0.utah.cloudlab.us.out
khmaja@ctl:~$
```

9. Checking the correct number of live nodes on 50070 site

CloudLab - Experiment Status

Namenode information

Not Secure | amd155.utah.cloudlab.us:50070/dfshealth... | All Bookmarks

Hadoop

Overview

Datanodes

Datanode Volume Failures

Snapshot

Startup Progress

Utilities

Overview

'ctl:9000' (active)

Started:	Sat Oct 07 01:27:12 UTC 2023
Version:	2.7.2, rb165c4fe8a74265c792ce23f546c64604acf0e41
Compiled:	2016-01-26T00:08Z by jenkins from (detached from b165c4f)
Cluster ID:	CID-47783971-909f-42c1-9fa6-c0ca28c1bde2
Block Pool ID:	BP-885852019-128.110.219.66-1696641966909

Summary

Summary

Security is off.

Safemode is off.

1 files and directories, 0 blocks = 1 total filesystem object(s).

Heap Memory used 228.62 MB of 674 MB Heap Memory. Max Heap Memory is 889 MB.

Non Heap Memory used 43.1 MB of 43.97 MB Committed Non Heap Memory. Max Non Heap Memory is -1 B.

Configured Capacity:	125.34 GB
DFS Used:	8 KB (0%)
Non DFS Used:	24 KB
DFS Remaining:	125.34 GB (100%)
Block Pool Used:	8 KB (0%)
DataNodes usages% (Min/Median/Max/stdDev):	0.00% / 0.00% / 0.00% / 0.00%
Live Nodes	2 (Decommissioned: 0)
Dead Nodes	0 (Decommissioned: 0)
Decommissioning Nodes	0
Total Datanode Volume Failures	0 (0 B)
Number of Under-Replicated Blocks	0
Number of Blocks Pending Deletion	0

Decommissioning Nodes	0
Total Datanode Volume Failures	0 (0 B)
Number of Under-Replicated Blocks	0
Number of Blocks Pending Deletion	0
Block Deletion Start Time	06/10/2023, 21:27:12

NameNode Journal Status

Current transaction ID: 3

Journal Manager	State
FileJournalManager(root=/dev/hdfs/dfs/name)	EditLogFileOutputStream(/dev/hdfs/dfs/name/current/edits_inproc)

NameNode Storage

Storage Directory	Type	State
/dev/hdfs/dfs/name	IMAGE_AND_EDITS	Active

Hadoop,
2015.

JPS command for each node

```
jai_mehta — khmaja@cp-2: ~ — ssh khmaja@c220g1-030606.wisc.cloudlab...

[khmaja@ctl1:~$ jps
205010 SecondaryNameNode
207483 Jps
102172 NameNode
[khmaja@ctl1:~$ ssh cp-1
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Wed Oct 11 15:22:15 2023 from 192.168.0.1
[khmaja@cp-1:~$ jps
58531 NodeManager
59859 Jps
50452 DataNode
58172 SecondaryNameNode
[khmaja@cp-1:~$ ssh cp-2
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
Last login: Wed Oct 11 15:22:53 2023 from 192.168.0.7
[khmaja@cp-2:~$ jps
48704 DataNode
56289 Jps
54571 SecondaryNameNode
54942 NodeManager
[khmaja@cp-2:~$
```